BrightLearn Exercise 3 : SQL Case statement

01)
```
SELECT product_name, price,
    CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
        WHEN price < 100 THEN 'Budget'
    END AS price_category
FROM products;
```

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products

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

02)
```
SELECT customer_name, amount,
    CASE
        WHEN amount >= 1000 THEN 'High Value'
        WHEN amount >= 500 AND amount < 1000 THEN 'Medium Value'
        WHEN amount < 500 THEN 'Low Value'
    END AS order_value_category
FROM orders;
```

orders ⇩

customer_name	amount	order_value_category
Alice	150,00	Low value
Bob	560,00	Medium value
Charlie	999,99	Medium value
Diana	45,50	Low value
Ethan	1200,00	High value

03) SELECT emp_name, department, salary
CASE
WHEN department = 'IT' AND salary > 8000 THEN 'Senior IT'
WHEN department = 'HR' AND salary > 5500 THEN 'Experienced HR'
ELSE 'Stuff'
END AS position_level
FROM employees ;

employees

emp_name	department	salary	position_level
John	IT	85 000	Senior IT
Sara	HR	60 000	Experienced HR
Mark	IT	75 000	Stuff
Lucy	Finance	95 000	Staff
Tom	HR	55 000	Staff

04) SELECT student name, score
CASE
WHEN score >= 90 THEN 'A'
WHEN score >= 80 THEN 'B'
WHEN score >= 70 THEN 'C'
WHEN score >= 60 THEN 'D'
ELSE 'F'
END AS grade
FROM students ;

student_name	score	grade
Ana	92	A
Ben	76	C
Cara	54	F
David	83	B
Ella	68	D

05) SELECT delivery_id, delivery_time_minutes,
CASE
WHEN delivery_time_minutes <= 30 THEN 'Fast'
WHEN delivery_time_minutes <= 60 THEN 'On Time'
ELSE 'Late'
END AS performance
FROM deliveries;

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deliveries

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Late
4	65	Late
5	100	Late

06) SELECT issue type, priority,
CASE
WHEN priority = 3 THEN 'High'
WHEN priority = 2 THEN 'Medium'
WHEN priority = 1 THEN 'Low'
END AS priority label
FROM tickets;

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tickets

issue type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

07) SELECT student_id,
(days_present * 100 / total days)
AS attendance_percentage,
CASE
WHEN (days_present * 100 / total_days) >=90
THEN 'Excellent'
WHEN (days_present * 100 / total days) >=75
THEN 'Good'
ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;

attendance ⇩

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improveit
5	100	Excellent

08) SELECT product_id, stock_qty,
CASE
WHEN stock_qty=0 THEN 'Out of stock'
WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
WHEN stock_qty >5 THEN 'In stock'
END AS stock_status
FROM products_inventory ;

08)

product_inventory

product_id	stock_qty	stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

09)
```
SELECT subject, enrolled students,
    CASE
        WHEN enrolled_students >= 25 THEN 'Large'
        WHEN enrolled_students >= 10 THEN 'Medium'
        ELSE 'Small'
    END AS class_size_category
FROM classes;
```

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classes

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10) SELECT payment_id, payment method, amount
CASE
WHEN payment method = 'Cash' AND amount >= 200
THEN ~~'Elig~~ 'Eligible for Discount'
ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;

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payments

payment_id	payment_method	amount	discount eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for discount
3	Card	150.00	Not Eligible
4	PayPal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount